Expert Feedback Cover Note

Dear Professor Harlow,

I have been working on a finite, executable benchmark package for observer-algebra diagnostics in static-patch-inspired regulator models. The basic setup compares a quantum matrix observer algebra with an abelian dephased control that has the same declared screen-visible data. The finite theorem package shows that screen shadows do not determine the observer algebra, while intrinsic operator response and cutoff-compatible continuity distinguish the two cases. It also includes a Type-`II_1` scaffold and a consecutive-cutoff embedding audit comparing a trace-filled UCP baseline with harmonic, heat-kernel, and Berezin-Toeplitz-inspired finite refinements.

A latest finite audit sharpens the embedding question: global scalar fuzzy-sphere low modes preserve screen data but their commutators vanish, whereas an observer-local coherent-state/tangent-plane scaling has a nonzero finite commutator lower bound on fixed low-excitation windows.

I am not claiming a continuum de Sitter theorem, a dS/CFT construction, or ER=EPR in de Sitter. The intended contribution is a finite diagnostic benchmark and a conditional lift obstruction: any proposed dictionary that factors only through screen-shadow data would be incomplete if the stated lift conditions hold.

The question where expert guidance would be most valuable is the cutoff embedding/coarse-graining problem. Are the continuum-lift obstruction and cutoff embedding problem meaningful ways to formalize the limitation of diagonal/screen-shadow data? If so, should the finite maps be organized through harmonic projection, heat-kernel coarse graining, Berezin-Toeplitz-style quantization, observer-local tangent-plane scaling, or another conditional expectation?

To keep the first pass short, I am attaching only the short feedback note. The TeX paper draft and executable certificate repository are available as supporting material.

I would be grateful for feedback on whether this is already known in the operator-algebra/QEC literature, and whether the lift/embedding formulation is worth formalizing further.

Expert Feedback Note: Finite Static-Patch Observer-Algebra Diagnostics

One-Paragraph Summary

We built a finite executable benchmark for a static-patch observer-algebra question: do screen-visible data determine the observer algebra? In the finite model, a quantum regulator with observer algebra `M_N` and a dephased abelian control with algebra `C^N` have identical diagonal/screen shadows, while intrinsic operator response distinguishes them by the commutator witness `[e_12,e_21]=e_11-e_22`. The benchmark includes finite derived dynamics, strong-continuity gates, a conditional Type-`II_1` scaffold, and an approximate consecutive-cutoff embedding audit comparing a trace-filled UCP baseline with harmonic, heat-kernel, and Berezin-Toeplitz-inspired refinements. The result is not a continuum de Sitter theorem. The purpose of this note is to ask which cutoff embedding or coarse-graining should be considered physically natural for static-patch observer algebras.

What Is Claimed

The finite claim is:

```
screen-shadow diagnostics do not determine finite observer algebras;
operator-response, continuity, and embedding data are necessary extra data.
```

The benchmark compares:

```
A_L = M_N,      S_L = C^N,      D_L = C^N.
```

Here `S_L` is the diagonal screen algebra and `D_L` is the dephased control. Any diagnostic that factors only through the diagonal screen data agrees between `A_L` and `D_L`. The algebras differ because `M_N` has nonzero commutators and `C^N` does not.

Theorem Stack

Screen-shadow collision. For diagnostics that factor through the diagonal screen map, the quantum and dephased controls have identical shadows. The operator-response witness separates them:

```
||[e_12,e_21]|| = 1 in M_N,       ||[x,y]|| = 0 in C^N.
```

Strong-continuity gate. If `Lambda_L(delta)=exp(delta G_L)`, `||G_L|| <= Gamma_L`, and `delta_L Gamma_L -> 0`, then

```
||Lambda_L(delta_L)-I|| <= exp(delta_L Gamma_L)-1 -> 0.
```

This excludes instantaneous dephasing without assuming off-diagonal survival.

Finite-to-Type-`II_1` scaffold. Exact full-matrix inclusions fail at consecutive spherical cutoffs `N_L=(L+1)^2`, but a cofinal factorial subsequence admits trace-preserving inclusions. The quantum path gives a UHF algebra whose tracial closure is the hyperfinite Type `II_1` factor by standard operator algebra facts; the dephased path has the same screen shadows and an abelian limit. This is a scaffold, not a canonical de Sitter static patch.

Approximate consecutive refinement. The exact-inclusion obstruction can be softened by replacing `*-inclusions` with UCP trace-preserving maps. The baseline map is: for `n <= m`,

```
Phi(A)=V A V^* + tau_n(A)(I_m - V V^*)
```

is unital, completely positive, and normalized-trace preserving. It is not multiplicative, but for `A=e_12`, `B=e_21`,

```
||Phi(AB)-Phi(A)Phi(B)|| = 1/n.
```

Thus consecutive spherical cutoffs can be related with vanishing multiplicativity error `1/N_L`.

The current audit also tests three more structured finite maps: harmonic mode-label refinement, harmonic refinement followed by heat-kernel Schur coarse graining, and a Berezin-Toeplitz-inspired CP smoothing surrogate. These are not claimed canonical, but they show the obstruction is not only an artifact of the factorial-subsequence scaffold.

Conditional continuum-lift obstruction. If finite regulator sequences lift to a continuum/static-patch setting while screen shadows converge and a response witness remains separated, then any dictionary that factors only through screen shadows is incomplete.

Observer-local tangent audit. A later finite audit checks the next hinge. Canonical scalar fuzzy-sphere low modes preserve screen data but their commutator scale vanishes. By contrast, for a fixed observer pole and fixed excitation cutoff `R`, the coherent-state tangent-plane scaling `a_L=J_+/sqrt(2j)` obeys

```
[a_L,a_L^*] >= (1 - 2R/L) I
```

on the low-excitation window. This is a finite observer-local theorem candidate, not a continuum static-patch theorem; it shifts the question to whether this tangent-plane sector is the right approximation to the observer algebra.

What Is Not Claimed

We do not claim:

- a continuum de Sitter observer-algebra theorem;
- a dS/CFT dictionary;
- literal ER=EPR in de Sitter;
- that the factorial subsequence is physically canonical;
- that the trace-filled UCP map is the correct fuzzy-sphere/static-patch

refinement;

- approximate QEC stability in a Type-II or Type-III limit.

Feedback Question

The current finite package isolates the remaining load-bearing physics choice:

```
Are the continuum-lift obstruction and cutoff embedding/coarse-graining problem
meaningful ways to formalize the limitation of diagonal/screen-shadow data?
```

Candidates include:

- angular-momentum branching or mode-label refinement;
- coherent-state or Berezin-Toeplitz symbol/quantization maps;
- observer-local coherent-state/tangent-plane scaling;
- heat-kernel coarse graining;

- continuum `L^2(S^2)` projection maps;
- approximate UCP embeddings rather than exact full-matrix inclusions;
- a modular/KMS-based conditional expectation intrinsic to the observer

algebra.

The concrete question is whether cutoff-compatible strong continuity plus conditional-expectation covariance is the right finite shadow of static-patch observer dynamics, or whether a different operator-algebraic axiom should be used.

Minimal Reproduction

```
PYTHONPATH=. python3 examples/reproduce_static_patch_package.py
PYTHONPATH=. python3 -m unittest tests.test_static_patch_strong_continuity tests.test_typeii_static_patch_
```

The paper-shaped note is:

```
paper/main.md
paper/main.tex
```